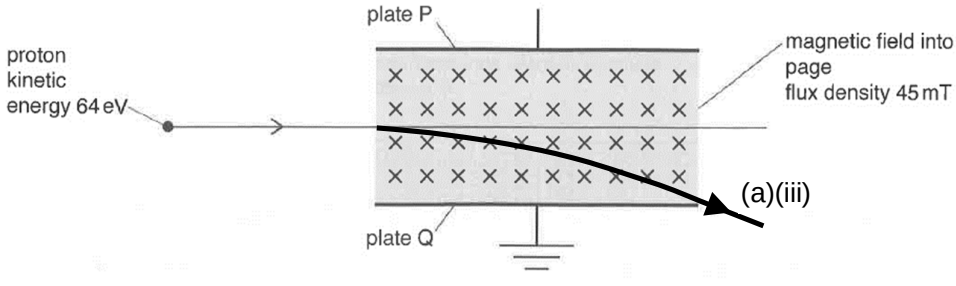


$$\approx 6.2 \times 10^{-19} \text{ J}$$

4(a)(i)	The magnetic force acting on the proton is vertically upwards as it enters the region between the parallel plates. In order to travel through the fields without deviating, the electric force acting on the proton has to be vertically downwards. The direction of the electric field must be vertically downwards, directed from plate P to plate Q. Hence, plate P has to be positive.
4(a)(ii)	Using $\frac{1}{2}m_p v^2 = E_k$, $v = \sqrt{\frac{2E_k}{m_p}}$ Proton not deviated as it passes between the plates, magnitude of magnetic force is equal to that of electric force on the proton. Using $Bev = eE$ $E = Bv$ $= B \sqrt{\frac{2E_k}{m_p}}$ $= (45 \times 10^{-3}) \sqrt{\frac{2(64)(1.60 \times 10^{-19})}{1.67 \times 10^{-27}}}$ $= 4983$ $\approx 5000 \text{ N C}^{-1}$
4(a)(iii)	With kinetic energy less than 64 eV, the magnitude of the magnetic force acting on the proton will be smaller than that of the electric force. The proton will experience a resultant force downwards, causing it to travel in a deviated path towards plate Q.

No.	Solution
	 The diagram shows a proton beam entering from the left, labeled 'proton kinetic energy 64 eV'. It passes between two parallel horizontal plates, 'plate P' (top) and 'plate Q' (bottom). Plate Q is connected to ground. A magnetic field, represented by 'x' marks, points into the page with a 'flux density 45 mT'. The beam is deflected downwards by the magnetic force, and the deflection is labeled '(a)(iii)'.
4(b)	Force per unit length on wire X, $\frac{F_x}{L_x} = \frac{\mu_0 I_Y I_X}{2\pi d}$ $= \frac{(4\pi \times 10^{-7})(8.4)(5.5)}{2\pi(0.12)}$ $= 7.7 \times 10^{-5} \text{ N m}^{-1}$ Direction: force on wire X is towards wire Y (right).
5(a)	E.m.f. of a source is the amount of energy converted from non-electrical to electrical per unit charge flowing through the source.